

# SHEHAN ROSHAN JAMAL

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## SUMMARY

Software engineer with 2+ years of professional full-stack development experience delivering solutions for international clients across Poland, the Netherlands, Australia and the US. I am currently completing a Master of Software Engineering in Auckland, where I have taken a geospatial research platform from requirements analysis and schema design through to production as its technical owner, and built an explainable-AI system from the ground up achieving 92% prediction accuracy. Strong across Python, TypeScript/React/Next.js, PostgreSQL and AWS, and comfortable owning a system end-to-end from design documentation to deployment.

## WORK HISTORY

### Software Engineering Intern | Tiaki Taonga Trust, New Zealand

April 2026 – Present (15-week internship) | Technical owner, Geo-Spatial & Policy Research Platform

- Took the platform from requirements through to production as its technical owner, authoring the three mentor-approved design documents Requirements Analysis, Database Schema, and API Design that gated the start of development.
- Defined and delivered all six functional and five non-functional requirements, covering interactive spatial visualization, cultural classification, document versioning, layer-based filtering, and spatial proximity search.
- Designed a PostgreSQL/PostGIS schema on a hub-and-spoke model that makes orphaned records structurally impossible, with foreign-key and CHECK constraints, indexes, and audit triggers producing an immutable change log.
- Built 25+ REST endpoints in TypeScript with a standardized response envelope, custom domain error codes, pagination, fully parameterized SQL, and a published OpenAPI 3.0 specification.
- Built the frontend in React/Next.js with Leaflet-based interactive mapping, permission-aware conditional rendering, and a 7-role RBAC system extended by a delegated per-project permission model.
- Migrated 19 API routes from Supabase PostgREST to raw parameterized SQL on AWS PostgreSQL, diagnosing spatial-function resolution, geometry serialization, and connection-pooling defects along the way.
- Led cross-portfolio authentication integration into a shared monorepo, solving an identity-federation problem between integer-keyed and UUID-keyed user models without modifying 25+ downstream endpoints.
- Set up CI/CD with GitHub Actions, wrote versioned idempotent SQL migrations submitted via pull request, and produced full handover documentation.

### Software Engineer | IFS, Sri Lanka

01/2022 – 01/2025

- Delivered customizations, reports and new functionality for enterprise customers across Poland, the Netherlands, Australia and the US, working across IFS versions from App09 through Cloud 24R1.
- Acted as functional advisor to Polish and Australian customers on PSO and related modules and authored technical specifications and time-estimation sheets for customers and other developers.
- Built customer-specific reports and edited XML configuration to add fields and functionality, proposing alternative approaches that improved customization efficiency.
- Managed version uplifts (App09→App10, App10→Cloud) and delivered application translations, including Polish and French.
- Built and maintained Tar scripts for environment data setup across application versions and ran internal presentations onboarding new employees onto core functionality.

### Intern | Agile Financials, Sri Lanka

06/2020 – 06/2021

- Built the full sign-up and login flow in React, integrated with Auth0, including email confirmation.
- Designed the PostgreSQL database and built the backend API and GraphQL layer in Golang.
- Wrote integration and unit tests for both the GraphQL API and the Golang backend.
- Sourced historical data from the Colombo StockExchange for the project's model and presented the finished project to National Trust Bank board members.

## EDUCATION

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**Master of Software Engineering** | *Yoobee College of Creative Innovation, New Zealand*

*Ongoing capstone: TFT-Predict v3, an explainable AI overlay for Teamfight Tactics*

**BEng (Hons) Software Engineering** | *Informatics Institute of Technology, Sri Lanka*

*Affiliated with the University of Westminster, UK*

**Foundation Certificate in Higher Studies** | *Informatics Institute of Technology, Sri Lanka*

## PERSONAL & ACADEMIC PROJECTS

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**TFT-Predict v3 MSE Capstone** | *Yoobee Colleges*

2026

- Built an explainable AI overlay for Teamfight Tactics using a screenshot-only pipeline: OCR via Tesseract/OpenCV feeding 23 engineered features into a CatBoost model, achieving 92.08% Top 4 accuracy and 0.9798 AUC.
- Used TreeSHAP to generate human-readable explanations for predictions, surfaced through a FastAPI backend and Electron desktop overlay.
- Diagnosed and fixed a trait-ID mismatch between Riot's display names and internal Set-17 IDs, lifting trait match rate from 19.8% to 61.0%.

**Novel-to-Comic Generator Pipeline** | *Personal Project*

2026

- Built an end-of-the-end pipeline that converts novel text into illustrated comic panels, deployed on Vast.ai.
- Used FLUX.1 Dev (fp8) for image generation, PuLID Flux II for character consistency, and Llama 3.1 8B (via Ollama) for scene parsing, orchestrated through ComfyUI.

**ReachOut** | *Personal Project*

2025

- Architected a full-stack volunteer platform, deployed with Docker, using a Python backend, React frontend and Apollo middleware.
- Integrated OpenAI and AssemblyAI for voice-to-text task creation, cutting task creation time from 3 minutes to 30 seconds.

**Emotion Analyzer for Articles Research Project** | *Personal Project*

01/2022 – 01/2023

- Designed a Chrome extension backed by a Python API and a hybrid BERT / sentiment-analysis / LSTM model to rate articles on an emotional scale.

## TECHNICAL SKILLS

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Proficiency: Python, JavaScript/TypeScript, React, Next.js, SQL, PostgreSQL, Git, HTML/CSS

Familiar: Golang, GraphQL, AWS, Docker, CI/CD (GitHub Actions), Supabase, FastAPI, Azure, Jira, PLSQL

Applied ML: CatBoost, scikit-learn, OCR (Tesseract/OpenCV), TreeSHAP, BERT/LSTM

Beginner: Figma, C++, PHP, Java

## CERTIFICATIONS, AWARDS & COMPETITIONS

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AWS Cloud Foundation and AWS Cloud Security Fundamentals AWS Academy

Cutting Edge 2018, Informatics Institute of Technology 1st Place, Project and 1st Place, Creativity

Google Hash Code 2017 global participant | Programming & Math's, TISC University of Westminster

## PUBLICATIONS

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"A Literature Review of Recommendation Systems", 2020 IEEE International Conference for Innovation and Technology (INOCON). [ieeexplore.ieee.org/document/9298450](https://ieeexplore.ieee.org/document/9298450)